RAW EOS 6D MARK II FIRMWARE 1.1.1 MAGIC LANTERN v79 ~1.5 HOURS

6D2 Fast-Card Test

Thanks for testing. The goal is to find which raw video formats a fast SD card can record **continuously** on the 6D Mark II with this build, and whether long takes are stable. Tick the boxes as you go and fill in the results; this form saves in any PDF reader that can fill forms (Acrobat Reader, Preview, Edge, Chrome). Nothing here touches Canon settings or factory data.

Firmware 1.1.1 only. If the camera ever locks up, pull the battery and keep any `CRASH*.LOG` / `ASSERT*.LOG` from the card root.

00 What you need

- [ ] A 6D Mark II on firmware 1.1.1, and the fast card (details below). If you have two of the same card, benchmark both (section 02) and use either for the rest.
- [ ] A lens with an **AF/MF switch**, set to **MF**. With the lens on AF, Canon refuses the 5x zoom that the bit-depth switch and crop mode use.
- [ ] About **70 GB free** for the ladder. The stability takes come after a fresh format.
- [ ] A computer with Python 3, for `cardcheck.py` (in the zip, under extras/tools).

Card brand and model	Size (GB)	Speed class	Camera serial (optional)

01 Install

- [ ] **ML already on this camera:** copy `autoexec.bin` and the `ML/` folder from the zip to the card root, replacing the old ones.
- [ ] **Or, first time on this camera:** format the card in the camera, copy `autoexec.bin`, `ML/` and `ML-SETUP.FIR` (all in the zip) to the card root, then Wrench menu → Firmware → update from the card, and restart.
- [ ] The **Trash** button opens ML. ML → Modules: enable `mlv_lite`, `bench`, `mlv_play` and `silent` (`mlv_play` needs `silent` loaded; leave Silent Picture itself OFF), then restart.
- [ ] Camera: Movie mode, **FHD 23.98p** (NTSC), M exposure, ISO 100–800, shutter 1/48, lens on **MF**.

On a Mac, run the checker from Terminal and avoid browsing the card in Finder: Finder's `._` files corrupted this card's directory once.

02 Card speed

- [ ] ML → Debug → Benchmarks → Card Benchmarks → **Quick R/W benchmark (1 min)**.

Card 1 write (MB/s)	Card 1 read (MB/s)	Card 2 write (MB/s)	Card 2 read (MB/s)

For comparison, a SanDisk Extreme Plus (U3) writes about 65 MB/s in this camera. Rule of thumb: a format runs continuously when its MB/s is at most about 90 % of the write speed. Mark the ladder rows you expect to fit before you start, if you like.

03 Setting a format

Movie tab → **RAW video** (the first entry) → SET.

- **Resolution**: the 1080p list goes up to 2048; in crop mode it goes up to 3072. Each remembers its own pick. The line shows the exact size that will record: check it matches the table.

• **Aspect ratio**: **1.90:1 (DCI)** is at the end of the list. Inside it, **If too tall** is *Shrink width* (default) or *Clamp height* (the crop rows use Clamp height).

• **Data format**: after changing it, **close the ML menu**. The view zooms in and out once. Reopen it: the entry reads **live**. If it reads *at 14* with "Lens AF blocks the zoom", set the lens to MF and close the menu again (Canon's MENU and back also works).

• **Crop mode** → *1:1 (5x zoom)*: close the menu and the view goes to 5x. Crop always records at 29.97 fps.

04 Data-rate ladder

Record each row for **up to 60 s**. Tick **cont** if it ran the full minute, otherwise write the second it stopped by itself. Within a block, once two rows in a row stop early, skip the rest of that block.

ROW	RES.	ASPECT	IF TOO TALL	FORMAT	RECORDS	MB/S	CONT	STOPPED AT (S)
1080P - 23.976 FPS								
A	1920	2.39:1	Shrink width	12-bit	1920×804	55.5	☐	
B	2048	2.39:1	Shrink width	12-bit	2048×856	63.0	☐	
C	2048	1.90:1 (DCI)	Shrink width	10-bit	2048×1080	66.3	☐	
D	2048	16:9	Shrink width	10-bit	2048×1152	70.7	☐	
E	1920	16:9	Shrink width	12-bit	1920×1080	74.6	☐	
F	2048	1.90:1 (DCI)	Shrink width	12-bit	2048×1080	79.5	☐	
G	2048	16:9	Shrink width	12-bit	2048×1152	84.8	☐	
H	1920	16:9	Shrink width	14-bit	1920×1080	87.0	☐	
I	2048	1.90:1 (DCI)	Shrink width	14-bit	2048×1080	92.8	☐	
CROP MODE - 29.97 FPS, LENS ON MF								
J	1920	2:1	Clamp height	10-bit	1920×926	66.6	☐	
K	1920	2:1	Clamp height	12-bit	1920×926	79.9	☐	
L	2560	2:1	Clamp height	10-bit	2560×926	88.8	☐	
M	3072	2:1	Clamp height	10-bit	3072×926	106.6	☐	

☐ **Then run the checker** (section 08) before formatting the card for section 05.

05 Stability

☐ Format the card in the camera.

☐ Fastest **1080p** row that ran cont: one **10-minute** take (written as .MLV, .M00, .M01 ... - keep them together).

☐ Same row: **five 30 s takes back to back**, each started within 3 s of the previous stop.

☐ Fastest **crop** row that ran cont: one **2-minute** take.

06 Scratch audio

☐ Movie tab → **Scratch audio** (the last entry) → Mode **Auto with RAW**. Record row A for about 20 s, clapping once near the start and once near the end.

- [ ] A .WAV named after the clip appeared in `DCIM/AUDIO`, and nothing new in the card root.

07 Playback (new in v79)

With RAW video on, press **PLAY** in movie LiveView: ML's player opens on the latest clip (the top dial switches clips, PLAY pauses). The top right reads `rd 41ms dr 30ms 12/240`: the time to **r**ead each frame from the card and to **d**raw it (INFO hides and shows it). **SET** opens the player's controls, where *color/fast* and *all/exact* are.

Play one clip of each depth from the ladder. Note whether there is a picture (10/12-bit clips showed nothing before v79), how the shadows look, and the rd / dr numbers while it plays.

- [ ] **14-bit** (row H or I)

Picture? (y/n)	Shadows (ok / crushed)	rd (ms)	dr (ms)	Clip

- [ ] **12-bit** (row A, B, E or F)

Picture? (y/n)	Shadows (ok / crushed)	rd (ms)	dr (ms)	Clip

- [ ] **10-bit** (row C, D or J)

Picture? (y/n)	Shadows (ok / crushed)	rd (ms)	dr (ms)	Clip

- [ ] Switch *color* / *fast* and *all* / *exact* in the player's controls: what changes (smoothness, rd/dr)?

08 Run the checker

With the card in the computer: `python cardcheck.py E:\` (macOS: `python3 cardcheck.py /Volumes/EOS_DIGITAL`). It only reads the card and saves `cardcheck-report-<date>-<time>.txt` in the current folder.

- [ ] Ran it after the ladder (section 04).
- [ ] Ran it again at the end.

09 Send back

This filled-in PDF, both `cardcheck-report-*.txt` files, and any `CRASH*` / `ASSERT*` logs. Clips are only needed if a report lists a **PROBLEM** for them.

Anything odd: warnings on screen, takes that stopped on their own, an overheating icon